



ASSIGNMENT

WAZUH Lab Setup

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WAZUH DEPLOYMENT AND HOST-BASED INTRUSION DETECTION LAB REPORT

1. Main Objectives

The primary objectives of this lab were:

- To deploy a centralized Wazuh Security Information and Event Management (SIEM) system.
- To configure Ubuntu Server as the Wazuh Manager.
- To install and configure Wazuh agents on multiple Linux endpoints.
- To establish secure communication between the manager and agents.
- To simulate attack scenarios including port scanning, brute-force attempts, and privilege escalation.
- To analyze detection results using Wazuh dashboards and MITRE ATT&CK mapping.
- To validate host-based intrusion detection capabilities in a controlled SOC environment.

2. Wazuh Manager Deployment (Ubuntu Server Setup)

2.1 Installation of Ubuntu Server

Ubuntu Server 22.04 LTS was installed as a virtual machine using VMware. The server was configured with host-only networking to allow communication within the lab environment.

Description: This step established the central monitoring server responsible for collecting and analyzing security logs from all endpoints.

2.2 System Preparation

System packages were updated to ensure compatibility and stability before installing Wazuh.

```
sudo apt update  
sudo apt upgrade -y
```

Description: Updating the system ensured that all dependencies and kernel components were current before deploying the Wazuh stack.

2.3 Wazuh Installation

The official Wazuh installation script was executed to deploy the Manager, Indexer, and Dashboard.

```
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh  
sudo bash wazuh-install.sh -a
```

```
kaiju@wazuh:~$ curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
kaiju@wazuh:~$ sudo bash wazuh-install.sh -a
12/02/2026 14:40:21 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.5
12/02/2026 14:40:21 INFO: Verbose logging redirected to /var/log/wazuh-install.log
12/02/2026 14:40:30 INFO: Wazuh web interface port will be 443.
```

This step installed the complete Wazuh stack, enabling centralized log analysis and visualization.

2.4 Service Verification

`sudo systemctl status wazuh-manager`

```
kaiju@wazuh:~$ sudo systemctl status wazuh-dashboard
• wazuh-dashboard.service - wazuh-dashboard
   Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; vendor preset: enabled)
   Active: active (running) since Thu 2026-02-12 14:49:32 UTC; 2min 32s ago
   Main PID: 50613 (node)
     Tasks: 11 (limit: 7001)
    Memory: 161.8M
       CPU: 9.288s
   CGroup: /system.slice/wazuh-dashboard.service
           └─50613 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --max-http-header-size=
```

Service verification confirmed that the manager was operational and ready to receive agent connections.

3. Wazuh Agent Deployment

3.1 Agent Setup – Kali Linux

Add Wazuh Repository and GPG Key:

`curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dearmor -o /usr/share/keyrings/wazuh.gpg`

```
ammar@ammar-virtual-machine:~$ curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --dearmor -o /usr/share/keyrings/wazuh.gpg
```

`echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg]`

```
ammar@ammar-virtual-machine:~$ echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main" | sudo tee /etc/apt/sources.list.d/wazuh.list
deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main
```

`https://packages.wazuh.com/4.x/apt/ stable main" | sudo tee /etc/apt/sources.list.d/wazuh.list`
`sudo apt update`

Install Agent:

`sudo apt install wazuh-agent -y`

```
(kali㉿kali)-[~]
└─$ sudo apt install wazuh-agent -y

Installing:
  wazuh-agent

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 2024
  Download size: 13.2 MB
  Space needed: 48.9 MB / 61.8 GB available
```

Configure Manager IP :

in `/var/ossec/etc/ossec.conf` as `192.168.153.138`

```
<ossec_config>
  <client>
    <server>
      <address>192.168.153.138</address>
      <port>1514</port>
```

Register Agent:

`sudo /var/ossec/bin/agent-auth -m 192.168.153.138`

```
(kali㉿kali)-[~]
└─$ sudo /var/ossec/bin/agent-auth -m 192.168.153.138

2026/02/12 11:21:33 agent-auth: INFO: Started (pid: 187627).
2026/02/12 11:21:33 agent-auth: INFO: Requesting a key from server: 192.168.153.138
2026/02/12 11:21:33 agent-auth: INFO: No authentication password provided
2026/02/12 11:21:33 agent-auth: INFO: Using agent name as: kali
2026/02/12 11:21:33 agent-auth: INFO: Waiting for server reply
2026/02/12 11:21:33 agent-auth: INFO: Valid key received
```

Start Agent:

`sudo systemctl enable wazuh-agent`

`sudo systemctl start wazuh-agent`

```
(kali㉿kali)-[~]
└─$ sudo systemctl enable wazuh-agent
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' → '/usr/lib/systemd/system/wazuh-agent.service'.

(kali㉿kali)-[~]
└─$ sudo systemctl start wazuh-agent
```

Kali was configured as both an attacker and monitored endpoint.

3.2 Agent Setup – Linux Mint (Main Victim)

Install Agent:

`sudo apt update`

`sudo apt install wazuh-agent -y`

```
ammar@ammar-virtual-machine:~$ sudo apt install wazuh-agent=4.7.5-1 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wazuh-agent
0 upgraded, 1 newly installed, 0 to remove and 102 not upgraded.
Need to get 9,379 kB of archives.
```

Configure Manager IP :

in `/var/ossec/etc/ossec.conf` as `192.168.153.138`

```
<ossec_config>
  <client>
    <server>
      <address>192.168.153.138</address>
      <port>1514</port>
```

Register Agent:

`sudo /var/ossec/bin/agent-auth -m 192.168.153.138`

```
ammar@ammar-virtual-machine:~$ sudo /var/ossec/bin/agent-auth -m 192.168.153.138
2026/02/13 00:11:53 agent-auth: INFO: Started (pid: 3495).
2026/02/13 00:11:53 agent-auth: INFO: Requesting a key from server: 192.168.153.138
2026/02/13 00:11:53 agent-auth: INFO: No authentication password provided
2026/02/13 00:11:53 agent-auth: INFO: Using agent name as: ammar-virtual-machine
2026/02/13 00:11:53 agent-auth: INFO: Waiting for server reply
2026/02/13 00:11:53 agent-auth: INFO: Valid key received
```

Start Agent:

`sudo systemctl enable wazuh-agent`

`sudo systemctl start wazuh-agent`

```
ammar@ammar-virtual-machine:~$ sudo systemctl enable wazuh-agent
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service → /usr/lib/systemd/system/wazuh-agent.service.
ammar@ammar-virtual-machine:~$ sudo systemctl start wazuh-agent
```

Mint served as the primary monitored victim system in the lab.

3.3 Agent Setup – Ubuntu

Install Agent:

`sudo apt update`

`sudo apt install wazuh-agent -y`

```
ammar@ammar-vmwarevirtualplatform:~/Desktop$ sudo apt install wazuh-agent=4.7.5-1 -y
Installing:
  wazuh-agent

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 194
  Download size: 9,379 kB
  Space needed: 31.5 MB / 14.4 GB available
```

Configure Manager IP:

in `/var/ossec/etc/ossec.conf` as 192.168.153.138

```
<ossec_config>
  <client>
    <server>
      <address>192.168.153.132</address>
      <port>1514</port>
```

Register Agent:

`sudo /var/ossec/bin/agent-auth -m 192.168.153.138`

```
ammar@ammar-vmwarevirtualplatform:~/Desktop$ sudo /var/ossec/bin/agent-auth -m 192.168.153.138
2026/02/13 01:27:19 agent-auth: INFO: Started (pid: 3194).
2026/02/13 01:27:19 agent-auth: INFO: Requesting a key from server: 192.168.153.138
2026/02/13 01:27:19 agent-auth: INFO: No authentication password provided
2026/02/13 01:27:19 agent-auth: INFO: Using agent name as: ammar-vmwarevirtualplatform
2026/02/13 01:27:19 agent-auth: INFO: Waiting for server reply
2026/02/13 01:27:19 agent-auth: INFO: Valid key received
```

Start Agent:

`sudo systemctl enable wazuh-agent`

`sudo systemctl start wazuh-agent`

```
ammar@ammar-vmwarevirtualplatform:~/Desktop$ sudo systemctl enable wazuh-agent
Created symlink '/etc/systemd/system/multi-user.target.wants/wazuh-agent.service' → '/usr/lib/systemd/system/wazuh-agent.service'.
ammar@ammar-vmwarevirtualplatform:~/Desktop$ sudo systemctl start wazuh-agent
```

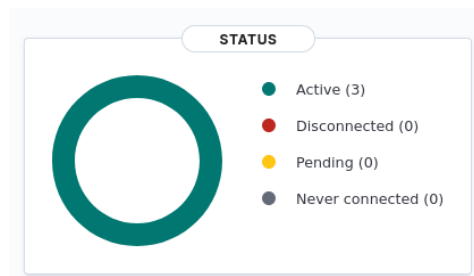
Lubuntu was configured as an additional monitored Linux endpoint.

4. Agents and Network Architecture

Agents (3)

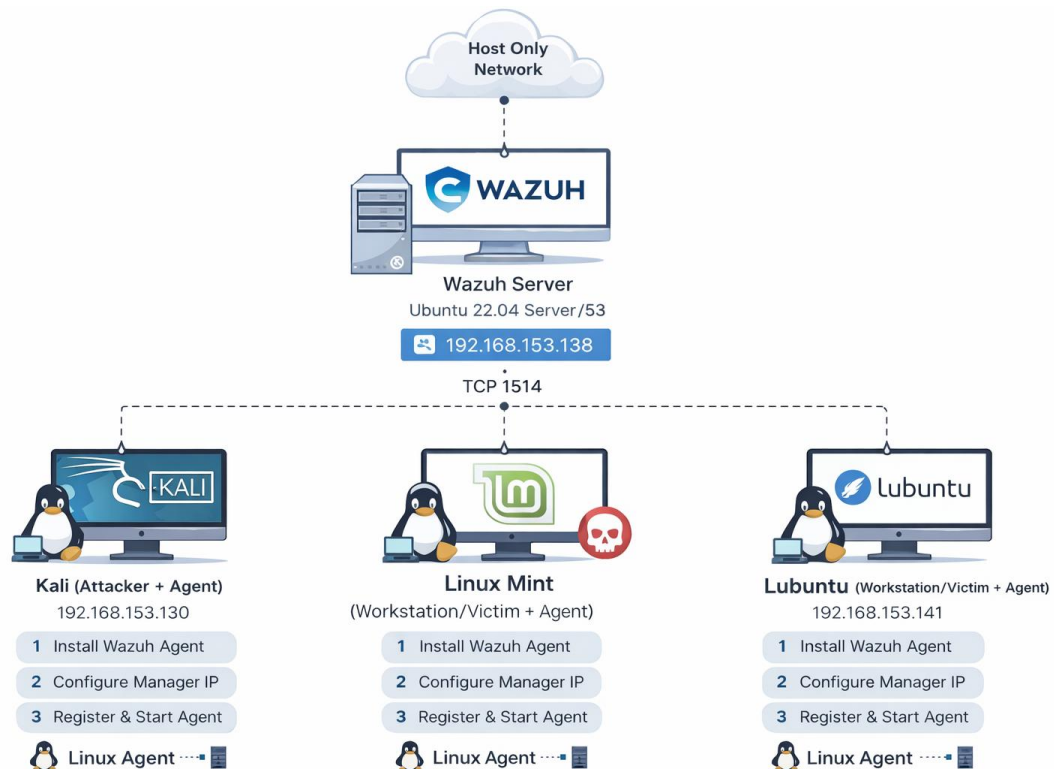
id!=000 and status=active				
ID ↑	Name	IP address	Group(s)	Operating system
001	kali	192.168.153.130	default	Kali GNU/Linux 2025.3
002	ammam-virtual-machine	192.168.153.139	default	Linux Mint 22.3
003	ammam-vmwarevirtualplatform	192.168.153.141	default	Ubuntu 25.10

STATUS



we can confirm here that all the agents are up and running.

Network Architecture



Here we can see the complete network of an ubuntu server and three agents.

5. Results and Attack Simulation

Nmap Scan:

```
nmap -sS -A 192.168.153.139
```

Results: SSH (Port 22) open, OpenSSH 9.6p1 detected, Linux OS fingerprinted, 999 closed ports.

```
(kali㉿kali)-[~]
$ nmap 192.168.153.139

Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-12 16:14 -0500
Nmap scan report for 192.168.153.139
Host is up (0.00044s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 00:0C:29:85:84:93 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.19 seconds
```

SSH Brute Force:

```
for i in {1..6}; do ssh fakeuser@192.168.153.139; done
```

Detected Rule ID 5710 (Non-existent user login), Rule ID 5503 (PAM login failure).

```
(kali㉿kali)-[~]
$ for i in {1..6}; do ssh fakeuser@192.168.153.139; done

The authenticity of host '192.168.153.139 (192.168.153.139)' can't be established.
ED25519 key fingerprint is SHA256:EyaPtQpXRmd12zgt7d2G76H0eMMA2gNCCqErsvvJ4xA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.153.139' (ED25519) to the list of known hosts.
fakeuser@192.168.153.139's password:
Permission denied, please try again.
fakeuser@192.168.153.139's password: █
```

Privilege Escalation:

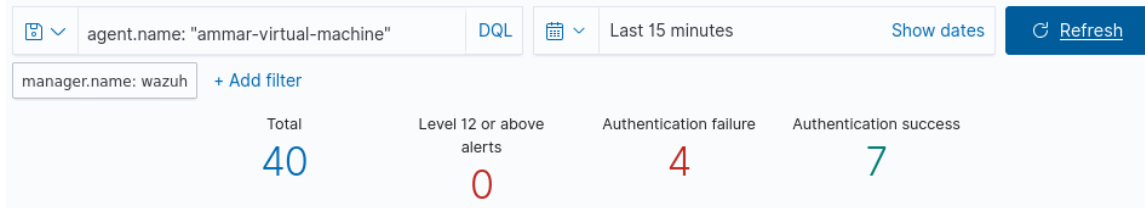
sudo su:

```
(kali㉿kali)-[~]
$ for i in {1..6}; do ssh fakeuser@192.168.153.139; done

The authenticity of host '192.168.153.139 (192.168.153.139)' can't be established.
ED25519 key fingerprint is SHA256:EyaPtQpXRmd12zgt7d2G76H0eMMA2gNCCqErsvvJ4xA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.153.139' (ED25519) to the list of known hosts.
fakeuser@192.168.153.139's password:
Permission denied, please try again.
fakeuser@192.168.153.139's password: █
```


Detected Rule ID 5402 (Successful sudo to ROOT).

ALERTS:



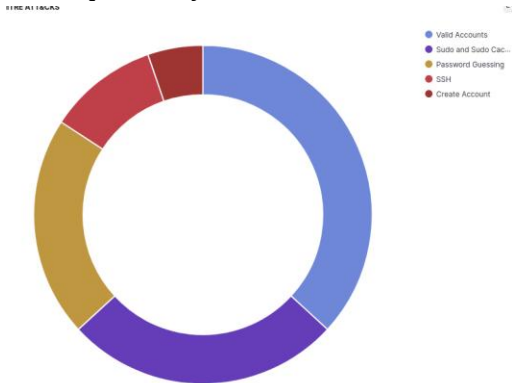
Alert Summary:

Total events: 40

Authentication failures: 4

Authentication successes: 7

Alerts primarily Level 3 and Level 5.



MITRE ATT&CK mappings observed:

- Credential Access
- Lateral Movement
- Privilege Escalation
- Defense Evasion

- Initial Access

Security Alerts 							
Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
> Feb 12, 2026 @ 16:16:26.23 1	002	ammar-virtual-machine	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
> Feb 12, 2026 @ 16:16:24.27 3	002	ammar-virtual-machine	T1110.001	Credential Access	PAM: User login failed.	5	5503
> Feb 12, 2026 @ 16:16:10.21 3	002	ammar-virtual-machine	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: Attempt to login using a non-existent user	5	5710
> Feb 12, 2026 @ 16:13:58.10 1	002	ammar-virtual-machine			PAM: Login session closed.	3	5502
> Feb 12, 2026 @ 16:13:58.09 1	002	ammar-virtual-machine	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	PAM: Login session opened.	3	5501
> Feb 12, 2026 @ 16:13:58.09 0	002	ammar-virtual-machine	T1548.003	Privilege Escalation, Defense Evasion	Successful sudo to ROOT executed.	3	5402

Conclusion: The Wazuh deployment successfully detected SSH brute-force attempts, authentication failures, privilege escalation, and mapped events to relevant MITRE ATT&CK techniques.